EAST WEST UNIVERSITY
Department of Computer Science and Engineering

CSE 475-Related work table

Course Name: Machine Learning

Course Code: CSE475

Section No: 06

Group No: 10

Dataset: Bot-IoT

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Submitted To:

Dr. Raihan Ul Islam
Associate Professor,
Department of Computer Science & Engineering .

Submitted By:

Name: Habibur Rahman
ID: 2023-1-60-024

Name: Tamim Rafi Ahmed

ID: 2021-3-60-089

Related work table

Authors(Year)	Title	Dataset	Application Area	Method/ model	Metrics	Strengths	Limitation/ gap	Relation to our work
1. Wai Weng Lo, Siamak Layeghy, Mohanad Sarhan, Marcus Gallagher, and Marius Portmann (2022)	E-GraphSAGE: A Graph Neural Network based Intrusion Detection System for IoT	BoT-IoT, ToN-IoT, NF-BoT-IoT, NF-ToN-IoT	Network Intrusion Detection Systems (NIDS) for Internet of Things (IoT)	**E-Graph SAGE** (An extended GraphSAGE architecture that incorporates edge flow features alongside topological information for	Accuracy, Precision, F1-Score, Detection Rate (Recall), False Alarm Rate (FAR), Weighted F1-Score	Leverages both topological graph structures and edge features directly for edge classification. Uses an inductive learning strategy, enabling it to classify	High computational complexity with full neighborhood aggregation on large-scale graphs. Does not currently utilize non-uniform or explainable	Demonstrates the effectiveness of GNN edge-embedding techniques over traditional flow-based network traffic independent models, serving as a primary benchmark

				edge classification via inductive representation learning)		unseen host nodes/IPs during deployment. Outperforms state-of-the-art methods across four comprehensive IoT benchmark datasets.	graph sampling models.	for GraphSAGE extensions in IoT intrusion detection.
2. Tien Ngo, Jiao Yin, Yong-Feng Ge, and Hua Wang (2025)	Optimizing IoT Intrusion Detection—A Graph Neural Network Approach with Attribute-Based Graph Construction	NF-BoT-IoT, NF-ToN-IoT	IoT Network Intrusion Detection Systems (NIDS) / Cybersecurity	**TKSGF** (Top-K Similarity Graph Framework using Cosine Similarity and FAISS) combined with **GraphSAGE**, **GCN**, and **GAT** architectures	Average F1-Score, Max F1-Score, Standard Deviation, Average Run Time	Replaces physical connection topology with attribute-based similarity graphs, resulting in more semantically meaningful node connections. Accelerates k-NN graph construction using GPU-optimized FAISS vector search. Provides extensive empirical analysis across graph directionality (directed vs.	GAT model performance severely degrades in scale and computation speed when handling large dynamic graphs. Heavy dependency on initial hyperparameter selection (such as similarity thresholds and top-k bounds)	Demonstrates that similarity-based topological graphs outperform direct physical link networks for learning flow representations, providing insights on hyperparameter selection and framework scaling for GNN-based IoT security systems.

						undirected), and GNN backbones.		
3. Yalu Wang, Zhijie Han, Jie Li, Xin He (2023)	BS-GAT: Behavior Similarity Based Graph Attention Network for Network Intrusion Detection	NF-BoT-IoT-v2 and NF-ToN-IoT-v2	IoT Network Intrusion Detection / Edge Security	**BS-GAT** (Behavior Similarity Graph Attention Network) — uses a 4-step hierarchical behavior matching rule (IPs, subnets, ports, prefixes) to construct edges and incorporates explicit behavioral weights into GAT self-attention.	Accuracy, Precision, Recall, F1-Score, and Gini Coefficient (neighbor distribution balance).	Prevents feature blurring and overfitting caused by node degree imbalance (achieves low Gini coefficients of 0.3529 and 0.4203). Maintains high multi-class detection stability across minority attack classes.	High graph construction time complexity on large-scale flow datasets.	Demonstrates how domain-specific behavior similarity rules can prevent degree imbalance in flow graphs, serving as a benchmark for custom edge-weighting strategies.
4. Wasswa et al. (2025/2026)	Graph Attention Neural Network for Botnet Detection: Evaluating Autoencoder, VAE and PCA-Based Dimen	CICIoT2022 and N-BaIoT	IoT Botnet Detection & Resource-Constrained Intrusion Detection	Combines a Variational Autoencoder (VAE) for latent space feature reduction (e.g., from 84/115 features down to 8 dimensions) followed by k-NN	Accuracy, Precision, Recall, F1-Score, Inference Latency, Model Storage Size.	Significantly reduces computational complexity and memory footprint by compressing high-dimensional tabular features prior to graph construction.	Latent feature compression leads to notable drops in recall for minority attack classes (e.g., brute force and flooding attacks).	Illustrates the computational trade-offs of latent feature compression prior to graph construction, underscoring the necessity of preserving fine-grained features for rare attack types.

	sion Reduction			graph generation and GAT classification.				
5. Riko Luša, Damir Pintar, and Mihaela Vranić	TE-G-SAGE: Explainable Edge-Aware Graph Neural Networks for Network Intrusion Detection	NF-UNSW-NB15-v3	Network Intrusion Detection Systems (NIDS) / Cybersecurity Anomaly & Threat Detection.	An inductive, edge-aware GraphSAGE-based model designed for edge classification on temporal flow graphs. Preserves strict chronological evaluation (60% train / 30% val / 10% test chronological splits) to eliminate data leakage.	Recall Precision, Recall-AUC / PR-AUC (suited for heavy class imbalance) F1-Score Feature Attribution & Interpretability Scores via SHAP values	Temporal Integrity: Enforces strictly chronological splits and temporal graph modeling, preventing future-data leakage common in random cross-validation. Relational Awareness: Leverages structural multi-hop connectivity (2-hop neighborhoods) alongside flow-level	Evaluation Scope: Primary empirical validation was conducted on a single main benchmark dataset (NF-UNSW-NB15-v3). Computational Overhead: Generating multi-hop computational subgraphs and SHAP explanations for every prediction adds processing latency in high-throug	TE-G-SAGE explicitly highlights the common pitfall of random train/test splits in flow-based NIDS, which introduce temporal data leakage. By adopting a time-respecting evaluation setup (chronological 60/30/10 split) on NetFlow traffic datasets like NF-UNSW-NB15-v3, TE-G-SAGE sets a

				Uses a adapted variable-level SHAP (SHapley Additive exPlanations) framework over two-hop computational subgraphs to explain predictions globally and locally.		statistical features. High Interpretability: Integrates SHAP directly into the edge-aware GNN computational graph, providing explicit feature attributions to support security analysts during incident triage. Inductive Capability: Generalizes to unseen network entities without requiring full graph retraining.	hput real-time environments. Static Snapshot Windows: Relies on structured temporal splits/batches rather than continuous real-time graph stream processing.	rigorous precedent for realistic deployment testing. Our work follows this methodology to ensure temporally valid performance metrics.
6. Ruisheng Li, Huimin Shen (Corresponding Author), and Qilong Zhang	A Dual-Perspective Self-Supervised IoT Intrusion Detection Method	ToN-IoT, NF-BoT-IoT, NF-CSE-CIC-IDS2018-v2, and NF-UNSW-NB15-v2	Network Intrusion Detection Systems (NIDS) / Cybersecurity in Internet of Things (IoT) environments	• Data Augmentation: Dual-perspective graph loss generation (Topology	F1	• Self-Supervised Representation: Learns rich structural and edge-level	Centralized Assumption: Evaluated primarily under centralized training, which may face latency and	Provides a strong reference architecture for lightweight, graph-based self-supervised learning on flow data, demonstrating how graph

	Based on Topolo gy Recon structio n and Featur e Perturb ation			Reconstr uction via edge sequence shuffling + Feature Perturbati on via edge attribute operation s like square/sq uare root) • Graph Encoder: EE-Graph SAGE (Edge-En hanced GraphSA GE for joint node-edg e aggregati on) • Loss Function: Adaptive Cosine Loss (ACos Loss) combinin g BCE with direction-aware cosine		represent ations without requiring massive labeled datasets. • Effective Edge Preservati on: Leverages flow-level metadata in edge features better than traditional GNNs that discard edge informatio n. • Lightweig ht & Fast: Low memory footprint and fast inference suitable for resource-c onstrained edge/IoT devices.	bandwidth constraints in real-world distributed edge topologies. • Static Snapshot Dependenc y: Graph construction relies on time-windo w flow aggregation , which might fail to capture fine-grained continuous temporal dynamic shifts. • Domain Portability: Highly customized augmentati on functions (like 20% feature perturbation) may require manual retuning for networks with vastly different protocol	contrastive/a ugmentation techniques can replace expensive supervised labeling while keeping models compact enough for real-time edge deployment

				similarity • Classifier: LightGBM downstream model		• Noise Robustness: Adaptive Cosine Loss mitigates label noise and balances confident vs. uncertain predictions.	characteristics.	

Table: Related work papers